

Properties of isosceles triangles

Two equal sides force two equal angles — and one line does the work of three.

LESSON 49–50

2 × 40 MIN

MATEMATIKA 7, §21

S8 5.2

LESSON CLOCK

15'

25'

25'

20'

5'

The base angles	15 min
The proof	25 min
The converse	25 min
One line, three roles	20 min
Homework	5 min

LEARNING OBJECTIVES

- State and prove that the base angles of an isosceles triangle are equal.
- State and use the converse.
- Know that the bisector from the apex is also the median and the altitude.
- Solve numerical problems on isosceles and equilateral triangles.

TEXTBOOK

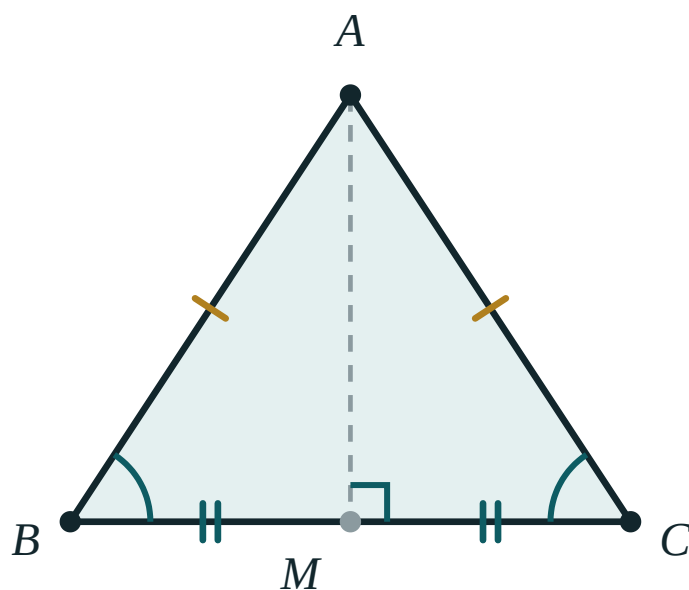
National	pp. 130–136
Cambridge	Stage 8, pp. 50–53
Workbook	Exercise 5.2

01

Explanation

The base angles

In an isosceles triangle the angles at the base are equal



$$AB = AC \implies \angle B = \angle C$$

Two equal sides, two equal base angles — and one axis of symmetry.

NAME	IN $\triangle ABC$ WITH $AB = AC$
lateral sides	AB, AC
base	BC
base angles	$\angle B, \angle C$
apex angle	$\angle A$

Since the angles add to 180° : $\angle B = \angle C = \frac{180^\circ - \angle A}{2}$, and $\angle A = 180^\circ - 2\angle B$.

ONE ANGLE DETERMINES ALL THREE

Give any one angle of an isosceles triangle — apex or base — and the other two follow at once. That is what makes these questions quick.

The proof

Draw the bisector AD of the apex angle.

STATEMENT	REASON
$AB = AC$	given
$\angle BAD = \angle CAD$	AD bisects $\angle A$
$AD = AD$	common
$\triangle ABD \equiv \triangle ACD$	first criterion
$\angle B = \angle C$	CPCT

THE AUXILIARY LINE IS THE WHOLE PROOF

Without AD there are no triangles to compare. Drawing the right extra line is the skill this proof teaches, and it recurs throughout the year.

The converse

If two angles of a triangle are equal, the sides opposite them are equal

So a triangle is isosceles **if and only if** two of its angles are equal. Either fact may be used to prove the other.

GIVEN	CONCLUDE
$AB = AC$	$\angle B = \angle C$
$\angle B = \angle C$	$AB = AC$
all three sides equal	all three angles 60°
all three angles equal	all three sides equal

A CONVERSE MUST BE PROVED SEPARATELY

That a statement is true does not make its converse true. Here both directions happen to hold, but each needs its own proof — and in general they can differ.

One line, three roles

In an isosceles triangle the bisector from the apex is also the median and the altitude to the base

It is also the axis of symmetry of the triangle. Four descriptions of one segment.

THE SEGMENT AD	BECAUSE
bisects $\angle A$	by construction
$BD = DC$	CPCT — it is a median
$AD \perp BC$	$\angle ADB = \angle ADC = 90^\circ$
is the axis of symmetry	the two halves are congruent

ONLY FROM THE APEX

The bisector from a **base** angle is not a median and not an altitude. The property belongs to the apex alone.

02 Worked examples

EXAMPLE 1 An isosceles triangle has apex angle 40° . Find the base angles.

1

The three angles add to 180° .

2

$180^\circ - 40^\circ = 140^\circ$ for the two base angles.

3

They are equal.

4

70° each.

ANSWER

70° each

EXAMPLE 2 An isosceles triangle has a base angle of 72° . Find the apex angle.

① Both base angles are 72° .

② $72 + 72 = 144$

③ $180 - 144$

④ $= 36^\circ$

ANSWER

36°

EXAMPLE 3 In $\triangle ABC$, $\angle B = \angle C = 55^\circ$ and $BC = 12$. What can be said about the sides, and about the altitude from A?

① Equal angles $\Rightarrow$ equal opposite sides: $AB = AC$.
The converse.

② So the triangle is isosceles with base BC .

③ The altitude from A is also the median.

④ It meets BC at its midpoint, 6 from each end.

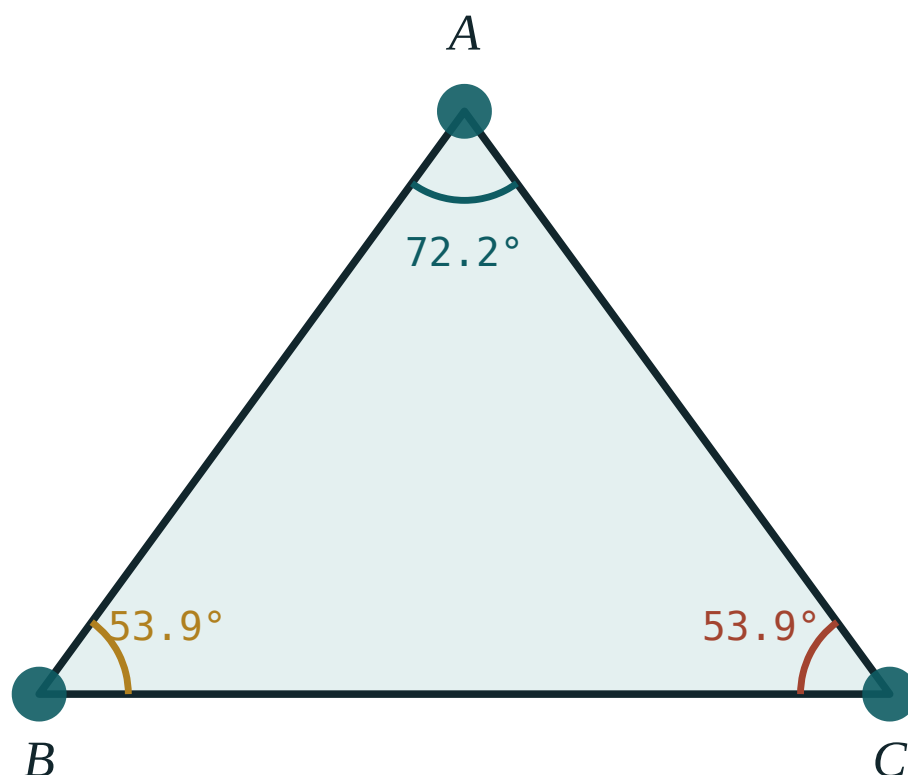
ANSWER

$AB = AC$; the altitude bisects BC

03 Interactive model

Fold a paper isosceles triangle along the line from the apex; the two halves coincide, and all four properties of that line appear at once.

Isosceles angles

 $\angle A$ 72.2° $\angle B$ 53.9° $\angle C$ 53.9° sum 180° longest side BC largest angle $\angle A$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Isosceles triangle	Teng yonli uchburchak	Равнобедренный треугольник
Lateral side	Yon tomon	Боковая сторона
Base	Asos	Основание
Base angle	Asosdagi burchak	Угол при основании
Apex angle	Uchidagi burchak	Угол при вершине
Converse	Teskari teorema	Обратная теорема
Axis of symmetry	Simmetriya o'qi	Ось симметрии
Property	Xossa	Свойство

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In an isosceles triangle the equal angles are:

at the apex

at the base

all three

none

Apex 40° gives base angles:

40°

70°

140°

100°

Base angle 72° gives apex:

36°

72°

108°

144°

Two equal angles imply:

nothing

two equal sides

a right angle

equal areas

The bisector from the apex is also:

a median only

an altitude only

both

neither

An equilateral triangle has angles:

45°

60°

90°

various

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Apex 40° : base angles

ANSWER 70°

2. Apex 90° : base angles

ANSWER 45°

3. Base angle 72° : apex

ANSWER 36°

4. Base angle 50° : apex

ANSWER 80°

5. Angles of an equilateral triangle

ANSWER 60°

6. Two equal angles imply

ANSWER Two equal sides

7. The bisector from the apex is also

ANSWER A median and an altitude

Medium · the standard question

7 PROBLEMS

1. Apex 100° : base angles

ANSWER 40°

2. Base angle 35° : apex

ANSWER 110°

3. An isosceles triangle with a 90° apex: base angles

ANSWER 45°

4. An isosceles triangle with a 90° base angle: possible?

ANSWER No

5. $\angle B = \angle C = 55^\circ$: $\angle A$

ANSWER 70°

6. Perimeter 26, base 10: the lateral sides

ANSWER 8 each

7. Perimeter 30, lateral side 11: the base

ANSWER 8

Hard · stretch and reason

7 PROBLEMS

1. An isosceles triangle with base 10 and lateral side 13: its height

ANSWER 12

2. Its area

ANSWER 60

3. An isosceles triangle in which the apex is twice a base angle

ANSWER $90^\circ, 45^\circ, 45^\circ$

4. An isosceles triangle in which a base angle is twice the apex

ANSWER $36^\circ, 72^\circ, 72^\circ$

5. An exterior angle at the base of an isosceles triangle is 130° : the apex

ANSWER 80°

6. Why can an isosceles triangle not have two obtuse angles?

ANSWER The sum would exceed 180°

7. A triangle with angles $x, x, 4x$

ANSWER $30^\circ, 30^\circ, 120^\circ$

07 Homework

Homework — 5 tasks

Say which angle is the apex and which are the base angles before computing.

1. An isosceles triangle has apex angle 64° . Find the base angles.
2. An isosceles triangle has a base angle of 43° . Find the apex angle.
3. An isosceles triangle has perimeter 34 and base 12. Find the lateral sides.
4. Prove that the base angles of an isosceles triangle are equal.
5. An isosceles triangle has base 16 and lateral side 17. Find its height and area.